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Deterministic symbolic regression with derivative information: General methodology and application to equations of state

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Abstract

Symbolic regression methods simultaneously determine the model functional form and the regression parameter values by generating expression trees. Symbolic regression can capture the complexity of real-world phenomena but the use of deterministic optimization for symbolic regression has been limited due to the complexity of the search space of existing formulations. We present a novel deterministic mixed-integer nonlinear programming formulation for symbolic regression that incorporates derivative constraints through auxiliary expression trees. By applying the chain rule to mathematical operations, binary expression trees are capable of representing the calculation of first and second derivatives. We apply this formulation to illustrative examples using derivative information to show increased model discrimination capability. Additionally, we perform a case study of a thermodynamic equation of state to gain insight on valid functional forms with thermodynamics-based constraints on the first and second derivatives.

KEYWORDS:
Surrogate modeling; Symbolic regression; Mixed-integer nonlinear programming; Cubic equations of state

INTRODUCTION

Traditional model regression methods restrict a model to fixed functional forms that are known before fitting the data. The pre-determined model structure aims to represent a response with known interactions of different variables based on prior knowledge. This fixed model structure restricts the scope of the regression to key parameters within its functional form, i.e., to constants appearing in linear, quadratic, logarithmic, exponential, or other set of functional terms. On the other hand, symbolic regression[1] defines the model structure while regressing coefficients allowing the relations of the variables to be discovered simultaneously with the parameter estimation.

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Symbolic regression allows the regression algorithm to flexibly define the functional form by representing a function as a set of operators or operands ($+,-,\div,\times$, $\exp(\cdot)$, $\log(\cdot)$,$(\cdot)^2$,$(\cdot)^3$, $\sqrt{(\cdot)}$, x, constant, etc.). By including the definition of the function form in the regression, model structure can be determined without prior knowledge of the relations of the variables and can provide insights into the underlying processes. These methods have been successful in a wide range of applications from determining governing equations to modeling dynamic robotic systems and simulating chemical processes and properties[2,3,4,5].

Symbolic regression methods depend on mathematical formulations being structured as an expression tree. A function is a recursive set of simple operations on values and the expression tree represents the order of operations or parenthetical structure. Operators are the mathematical procedures, such as addition or exponentiation, that act on the values of other tree nodes. Operands, constants and variables, are the numerical inputs from data or constant values that are leaves of the expression tree and do not act on other nodes. We focus on binary expression trees since most mathematical operators operate on one or two values.

Genetic Programming (GP) has been the traditional way to solve symbolic regression[1]. This approach to symbolic regression remains an active field of research[6,7,8] and there exist commercially available software applying GP, such as Eureqa[9]. The GP community is actively focused on improving accuracy of the symbolic models and efficiency of the regression algorithms to improve confidence in these techniques[10]. Recent developments[1,6,7,8] have focused on incorporating derivatives in fitness metrics or localized hybrid models for addressing more complex problems and improving previous search algorithms.

Kaizen Programming (KP) approaches symbolic regression using an algorithm structured in a Plan-Do-Check-Act methodology[11]. KP introduces the idea of multiple experts contributing to a solution and quantifying the contribution of subcomponents to the final model with statistics[12]. In a comparison of GP and KP, Ferreira et al.[13] showed that KP has better performance than GP for a bioethanol distillation column.

A recent physics inspired method for symbolic regression, AI Feynman[2], relies on brute-force search in the space of functions but includes additional information, such as units for physical models, and exploits quick regressions of neural networks to evaluate the relation of inputs to facilitate finding a symbolic expression. Udrescu and Tegmark[2] use insights from physics models such as the abundance of accurate low-order polynomial models and separability of variables, as well as unit-based dimensional analysis limitations between inputs and outputs. The method further controls the regression by incorporating neural networks to test for translational symmetry, separability, and further generalizations to combine inputs into intermediate expressions to find a symbolic equation. To improve the interpretability

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and reliability of the symbolic approximations, AI Feynman 2.0[14] evaluates graph modularity, compositionality and generalized symmetry, based on gradients to acquire simple, but accurate equations.

Recently, deterministic optimization algorithms have been proposed for symbolic regression. Cozad et al.[15,16] developed the first deterministic mixed-integer nonlinear programming (MINLP) formulation for the global optimization of symbolic regression. Building upon this structure, additional mathematical operators such as identity and null were added to speed up the regression and reduce the model size[17,18]. Neumann et al.[17] focused on modifications to the binary variable decisions to reduce the size of the MINLP by restricting constants and variables to leaf nodes and providing an identity operation. Kim et al.[19] developed a sequential tree construction heuristic called STreCH and proposed constraints to strengthen the formulation of Cozad et al. by eliminating redundant math operations from the feasible space of the MINLP. Austel et al.[20] developed a heuristic that speeds up the optimization by enumerating possible expression trees that include monomials of variables as input, thus allowing the formation of more complex functions with smaller trees at the expense of not guaranteeing global optimality. Wei et al.[18] developed a generalized Benders decomposition global optimization approach to solve symbolic regression using linear relaxations of the MINLP. These formulations have been applied to real-world problems from viscosity and kinetics models to physics-based models from the Feynman Database for Symbolic Regression[2,17].

Our work focuses on including derivative values in the regression as data points or constrained values to be regressed simultaneously with the main function. By providing derivative information, the regression method reduces the search space and guides the model selection based on the gradient for similar objective values of the primary functional expression. To incorporate derivatives in the model, we apply forward differentiation rules to the binary expression tree. Including derivative information in the regression can improve the gradients estimated for physical models thus constraining the regression to functions that are more physically meaningful.

In this paper, we formulate symbolic regression as a nonlinear nonconvex disjunctive program that can be solved with global optimization techniques. We formulate first and second derivative expression trees that are defined in parallel with the functional structure expression tree and apply constraints to the derivatives to control the determined structure. In addition, we strengthen the formulation with additional redundancy constraints to shrink the search space by reducing trivial solutions and with constraints on the variance of intermediate values of the mathematical operators to reduce unnecessary complexity. We demonstrate this formulation on a small example problem and a constrained regression of the alpha function that is embedded in the Peng-Robinson cubic equation of state.

The remainder of the paper is organized as follows. The next section reviews the primary MINLP model for symbolic regression and presents our proposed MINLP model for incorporating first and second derivatives. The following

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section provides additional constraints to apply to the tree structure to penalize model complexity and further reduce redundant operations. Subsequently, we apply the formulation to two illustrative example problems to demonstrate model discrimination due to constrained gradients and to demonstrate constrained extrapolated behavior. Then, we present an application to the embedded alpha function of cubic equations of state and evaluates the impact of the proposed methodology in the context of this application. Finally we provided conclusions in the last section.

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PROPOSED MODEL

In the next subsection, we review the nonconvex nonlinear disjunctive programming formulation presented by Cozad and Sahinidis [16], along with constraints from Kim et al. [19] that strengthen the Cozad-Sahinidis formulation. This MINLP facilitates the solution of the symbolic regression problem to global optimality and serves as the starting point of the model we propose to model derivatives in the subsection entitled "Auxiliary derivative expression tree formulation."

Primary expression tree formulation

The formulation represents a binary expression tree of depth D with $N = 2^D - 1$ nodes as an MINLP. At each parent node, an operand (variable or constant) or an operator (e.g., $+, -, \times, \div$) can be chosen and evaluated based on the values of the node's left and right child node. Mathematical operators are recursively applied to children nodes to calculate a value for the parent node. A node is active if it is assigned an operator or operand. Binary expression trees can be defined to a certain depth but may make use of fewer nodes than defined and have inactive nodes. Each parent node represents a disjunction, i.e., a decision that applies a different set of constraints on the problem. As a result, although a large number of equations are specified at every node, only a small portion of equations will be active at the final solution. A list of common operators, operands, and sets are listed in Table 1. Using this notation, we can model a binary expression tree as an MINLP and formulate the following nonconvex disjunctive program.

$$\min \sum_{i=1}^{n_{data}} (z_i - v_{i1})^2 \tag{1a}$$

$$\text{s.t.} \bigvee_{o \in \mathcal{O}} \begin{bmatrix} y_n^o = 1 \\ f_i^o(v_{in}, v_{il_n}, v_{ir_n}, x_{id}) = 0 \\ g_i^o(v_{in}, v_{il_n}, v_{ir_n}, x_{id}) \le 0 \\ i = 1, ..., n_{data} \end{bmatrix} \bigvee \begin{bmatrix} \sum_{o \in \mathcal{O}} y_n^o = 0 \\ v_{in} = 0 \\ i = 1, ..., n_{data} \end{bmatrix} \qquad n \notin \mathcal{T} \tag{1b}$$

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$$\bigvee_{o\in\mathcal{L}}\begin{bmatrix} y_n^o = 1 \\ f_i^o(v_{in}, x_{id}) = 0 \\ g_i^o(v_{in}, x_{id}) \le 0 \\ i = 1, ..., n_{data} \end{bmatrix} \bigvee \begin{bmatrix} \sum_{o\in\mathcal{L}} y_n^o = 0 \\ v_{in} = 0 \\ i = 1, ..., n_{data} \end{bmatrix} \qquad n \in \mathcal{T} \quad (1c)$$

$$\sum_{o\in\mathcal{O}} y_n^o \le 1 \qquad n = 1, ..., N \quad (1d)$$

$$\sum_{o\in\mathcal{B}\cup\mathcal{U}} y_n^o = \sum_{o\in\mathcal{O}} y_l^o \qquad n \notin \mathcal{T} \quad (1e)$$

$$\sum_{o\in\mathcal{B}} y_n^o = \sum_{o\in\mathcal{O}} y_r^o \qquad n \notin \mathcal{T} \quad (1f)$$

$$\sum_{o\in\mathcal{V}} \sum_{n=1}^{N} y_n^o \ge 1 \quad (1g)$$

$$y_n^o \in \{0, 1\} \qquad n \in \mathcal{T} \quad o \in \mathcal{L} \quad (1h)$$

$$v_{in} \in \left[v_{in}^{lo}, v_{in}^{up}\right] \qquad n = 1, ..., N \quad i = 1, ..., n_{data} \quad (1i)$$

This formulation minimizes the sum of squared errors between measurements z_i and model predictions v_{i1}, over the training data set $i = 1, ..., n_{data}$. The model is a function represented by the binary expression tree of predictor or input variables x_d with n dimensions represented by the tree nodes. The objective can be easily exchanged for other loss functions such as linear error or absolute deviation.

The disjunction for each node for $o \in \mathcal{O}$ is represented by y_n^o and defines what operand or operator is active at node n. If active ($y_n^o = 1$), the corresponding operation ${f_i}^o$ and necessary restrictions ${g_i}^o$ are enforced with common operations listed in Table 2. Restrictions include not allowing divisions by zero or logarithms operating on non-positive values. Constraint (1b) represents the disjunctions over the non-terminal nodes for which all operators can be chosen. Constraint (1c) represents terminal node disjunctions which only allow the leaf operators variables and constants. The logic constraints for the expression tree restrict the y_n^o decisions to one per node with Constraint (1d). There are three types of operands and operators, binary, unary, and leaves. A binary operand, such as +,−,×,÷, acts on the left and right children, while a unary operand, such as exp or log, acts only on the left child. A leaf node has either a constant or a predictor variable. Constraints (1e)–(1f) define the tree structure and restrict the allowed active node choices based on the parent node being active and child node activity. The bulk of the model comes from the Cozad

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et al. formulation, with the exception of the tree defining constraints (1e)–(1f) that were provided by the Kim et al. formulation. Constraint (1g) ensures at least one predictor variable is used as an input in the model.

We model the nonconvex nonlinear disjunctive program using a big-M MINLP formulation[21]. In the formulation, big-M values are determined using interval arithmetic to find the upper bound $\overline{M}^o_{in}$ and lower bound $\underline{M}^o_{in}$ for each constraint f^o_i at each node given limits on possible values of nodes v^{lo}_{in} and v^{up}_{in}. We define a tolerance, ϵ, for the math operator domain constraints, g^o_i, that enforce positive or non-zero inputs to the math operator. The optimization formulation is as follows:

Model *P*

$$\min \sum_{i=1}^{n_{data}} (z_i - v_{i1})^2$$

$$\text{s.t.} f^o_i(v_{in}, v_{il_n}, v_{ir_n}, x_{id}) \le \overline{M}^o_{in}(1 - y^o_n) \qquad n \notin \mathcal{T}, o \in \mathcal{O}, i = 1, ..., n_{data} \quad (2a)$$

$$f^o_i(v_{in}, v_{il_n}, v_{ir_n}, x_{id}) \ge \underline{M}^o_{in}(1 - y^o_n) \qquad n \notin \mathcal{T}, o \in \mathcal{O}, i = 1, ..., n_{data} \quad (2b)$$

$$g^o_i(v_{in}, v_{il_n}, v_{ir_n}, x_{id}) \le G^o_{in}(1 - y^o_n) \qquad n \notin \mathcal{T}, o \in \mathcal{O}, i = 1, ..., n_{data} \quad (2c)$$

$$v_{in} \le v^{up}_{in} \sum_{o \in \mathcal{O}} y^o_n \qquad n \notin \mathcal{T}, o \in \mathcal{O}, i = 1, ..., n_{data} \quad (2d)$$

$$v_{in} \ge v^{lo}_{in} \sum_{o \in \mathcal{O}} y^o_n \qquad n \notin \mathcal{T}, o \in \mathcal{O}, i = 1, ..., n_{data} \quad (2e)$$

$$f^o_i(v_{in}, x_{id}) \le \overline{M}^o_{in}(1 - y^o_n) \qquad n \in \mathcal{T}, o \in \mathcal{L}, i = 1, ..., n_{data} \quad (2f)$$

$$f^o_i(v_{in}, x_{id}) \ge \underline{M}^o_{in}(1 - y^o_n) \qquad n \in \mathcal{T}, o \in \mathcal{L}, i = 1, ..., n_{data} \quad (2g)$$

$$g^o_i(v_{in}, x_{id}) \le G^o_{in}(1 - y^o_n) \qquad n \in \mathcal{T}, o \in \mathcal{L}, i = 1, ..., n_{data} \quad (2h)$$

$$v_{in} \le v^{up}_{in} \sum_{o \in \mathcal{L}} y^o_n \qquad n \in \mathcal{T}, o \in \mathcal{L}, i = 1, ..., n_{data} \quad (2i)$$

$$v_{in} \ge v^{lo}_{in} \sum_{o \in \mathcal{L}} y^o_n \qquad n \in \mathcal{T}, o \in \mathcal{L}, i = 1, ..., n_{data} \quad (2j)$$

$$\sum_{o \in \mathcal{O}} y^o_n \le 1 \qquad n = 1, ..., N \quad (2k)$$

$$\sum_{o \in \mathcal{B} \cup \mathcal{U}} y^o_n = \sum_{o \in \mathcal{O}} y^o_r \qquad n \notin \mathcal{T} \quad (2l)$$

$$\sum_{o \in \mathcal{B}} y^o_n = \sum_{o \in \mathcal{O}} y^o_l \qquad n \notin \mathcal{T} \quad (2m)$$

$$\sum_{o \in \mathcal{V}} \sum_{n=1}^{N} y^o_n \ge 1 \quad (2n)$$

$$y^o_n \in \{0, 1\} \qquad n \notin \mathcal{T} \quad o \in \mathcal{O}$$

$$y^o_n \in \{0, 1\} \qquad n \in \mathcal{T} \quad o \in \mathcal{L}$$

$$v_{in} \in \left[v^{lo}_{in}, v^{up}_{in}\right] \qquad n = 1, ..., N \quad i = 1, ..., n_{data}$$

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Auxiliary derivative expression tree formulation

To include derivatives in the MINLP approach for symbolic regression, we define auxiliary binary expression trees that at the root node calculate the first and second derivatives of the response surface with respect to a single predictor variable x_d. The parallel expression trees are connected by the binary decision variables for choosing operators in the nodes of the main tree. Corresponding to the values v_{in} of expressions in the primary modeling expression tree, the first and second derivative node values are defined as v'_{in} and v''_{in}, respectively.

Similar to the calculation of the primary expression modeled value, operators are recursively applied on the children nodes from the root node down to the leaves building the expression for the first or second derivative value. The auxiliary binary expression trees use intermediate calculated values in the primary binary expression tree and apply the chain rule of the various operands to children nodes to calculate the derivatives at each node. An example of the auxiliary binary expression trees evaluations is shown in Figure 1.

The evaluation of first derivatives requires one application of the chain rule to the operand or operator. The expansion of the notation to first derivatives of common operands is shown in Table 3. The simplest case is addition or subtraction, which is calculated using the addition or subtraction of the first derivative of the children node. For multiplication and division using the chain rule, the intermediate value calculated at the node in the main expression tree is needed as is the case for all of the unary operations (exp, log, square, square root, etc.). The tolerances for forcing non-negative evaluations of square roots and blocking division by zero are still imposed on the derivative evaluations. The symbolic regression formulation can be modified as follows to include calculations for the derivative values at each node, and constraints on the first derivatives:

Model (P')

$$\min \sum_{i=1}^{n_{data}} (z_i - v_{i1})^2$$

s.t. Primary expression tree formulation

$$f'^{o}_{i}(v_{in}, v_{il_n}, v_{ir_n}, v'_{in}, v'_{il_n}, v'_{ir_n}, x_{id}) \le \overline{M'}^{o}_{in}(1 - y^o_n) \qquad n \notin \mathcal{T}, o \in \mathcal{O}, i = 1, ..., n_{data} \quad (3a)$$

$$f'^{o}_{i}(v_{in}, v_{il_n}, v_{ir_n}, v'_{in}, v'_{il_n}, v'_{ir_n}, x_{id}) \ge \underline{M'}^{o}_{in}(1 - y^o_n) \qquad n \notin \mathcal{T}, o \in \mathcal{O}, i = 1, ..., n_{data} \quad (3b)$$

$$f'^{o}_{i}(v_{in}, v'_{in}, x_{id}) \le \overline{M'}^{o}_{in}(1 - y^o_n) \qquad n \in \mathcal{T}, o \in \mathcal{L}, i = 1, ..., n_{data} \quad (3c)$$

$$f'^{o}_{i}(v_{in}, v'_{in}, x_{id}) \ge \underline{M'}^{o}_{in}(1 - y^o_n) \qquad n \in \mathcal{T}, o \in \mathcal{L}, i = 1, ..., n_{data} \quad (3d)$$

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$$h'_i(v_{in}, v_{il_n}, v_{ir_n}, v'_{in}, v'_{il_n}, v'_{ir_n}, x_{id}) \leq H'_{in} \qquad n = 1, ..., N, i = 1, ..., n_{data} \quad (3e)$$

$$v'_{in} \in \left[v'^{lo}_{in}, v'^{up}_{in}\right] \qquad n = 1, ..., N, i = 1, ..., n_{data}$$

where f'^{o}_i enforces the derivative of the selected math operator on a node with a Big-M constraint and h'_i is an optional constraint on the first derivative usually applied to the root node, where $n = 1$.

Second derivatives are calculated using the chain rule twice and require intermediate calculated values from both the primary and first derivative auxiliary expression trees to be computed. Applying the chain rule twice still relies on simple operations, but in general requires careful implementation due to the increased complexity in the function evaluations. The second derivative notation of common operands is shown in Table 4. Similar to the first derivative calculation shown in Figure 1, the calculation of second derivatives depends on values from the primary and first derivative auxiliary trees and is not an independent evaluation. The leaf nodes of the auxiliary second derivative tree are zeros for both constants and variables. If the input variable is a function of another variable, such as $x = f(T)$, input specific first and second derivative evaluations with respect to the variable T can be included in the formulation. Similar to including the first derivatives, the formulation can be modified to include second derivatives as follows:

Model (P'')

$$\min \sum_{i=1}^{n_{data}} (z_i - v_{i1})^2$$

s.t. Primary expression tree formulation

First derivative expression tree formulation

$$f''^{o}_i(v_{in}, v_{il_n}, v_{ir_n}, v'_{in}, v'_{il_n}, v'_{ir_n}, v''_{in}, v''_{il_n}, v''_{ir_n}, x_{id}) \leq \overline{M''}^{o}_{in}(1 - y^o_n) \qquad n \notin \mathcal{T}, o \in \mathcal{O}, i = 1, ..., n_{data} \quad (4a)$$

$$f''^{o}_i(v_{in}, v_{il_n}, v_{ir_n}, v'_{in}, v'_{il_n}, v'_{ir_n}, v''_{in}, v''_{il_n}, v''_{ir_n}, x_{id}) \geq \underline{M''}^{o}_{in}(1 - y^o_n) \qquad n \notin \mathcal{T}, o \in \mathcal{O}, i = 1, ..., n_{data} \quad (4b)$$

$$f''^{o}_i(v_{in}, v'_{in}, v''_{in}, x_{id}) \leq \overline{M''}^{o}_{in}(1 - y^o_n) \qquad n \in \mathcal{T}, o \in \mathcal{L}, i = 1, ..., n_{data} \quad (4c)$$

$$f''^{o}_i(v_{in}, v'_{in}, v''_{in}, x_{id}) \geq \underline{M''}^{o}_{in}(1 - y^o_n) \qquad n \in \mathcal{T}, o \in \mathcal{L}, i = 1, ..., n_{data} \quad (4d)$$

$$h''_i(v_{in}, v_{il_n}, v_{ir_n}, v'_{in}, v'_{il_n}, v'_{ir_n}, v''_{in}, v''_{il_n}, v''_{ir_n}, x_{id}) \leq H''_{in} \qquad n = 1, ..., N, i = 1, ..., n_{data} \quad (4e)$$

$$v''_{in} \in \left[v''^{lo}_{in}, v''^{up}_{in}\right] \qquad n = 1, ..., N, i = 1, ..., n_{data}$$

where f''^{o}_i enforces the second derivative of the selected math operator on a node with a Big-M constraint and h''_i is an optional constraint on the second derivative.

The inclusion of derivatives into the formulation creates an auxiliary binary tree for each derivative taken with respect to a variable. Additional big-M values can be calculated for the derivative values or a large bound can be applied

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to each operator in the derivative calculation. The disjunctions formulated by this expansion remain the same with additional equations applied to each disjunction for the evaluation of the derivatives. The binary variables determining the operator of a node remain the same between all trees, expanding the decision of an operator or operand at each node accounting for the calculated value to that of the derivative values. Overall, for each derivative with respect to a variable evaluated for an expression tree, an additional evaluated value for each node, n, and each observed data point, i, is added to the formulation, i.e., the values calculated now are v_{in}, v'_{in}, and v''_{in}. Taking the derivative with respect to multiple variables requires additional auxiliary binary trees enforcing the correct derivative values at the variable leaf nodes for applying the chain rule.

STRENGTHENING THE FORMULATION

The binary expression tree is a very flexible structure for representing functional forms. Changing the operator of a parent node near the top of the tree can produce very different mathematical expressions. However, with this flexibility comes equivalent representation of the same equation, such as from the commutative properties of multiplication or addition. To reduce the search space for the MINLP formulation, we add constraints to reduce redundant functional forms, prioritize nodes close to the root node, and break symmetry. We add these additional constraints to further reduce the search space and improve the speed of the regression. A summary of the constraints are listed in Table 5.

One class of constraints that reduce the search space targets trivial operations or evaluations. This is similar to the redundancy constraints that focus on manipulations of constant values or inverse operations such as a logarithm operating on an exponential operator. The set of constraints we introduce focuses on binary operators operating on the same input variable. For example, $x_1 - x_1$ or x_1/x_1 result in trivial solution of 0 or 1. For the operators $+$ or $*$ acting on the same variable, we restrict redundant expressions to be represented instead by $2x_1$ or the $(\cdot)^2$ operator.

A major issue with symbolic regression that leads to slow convergence of the MINLP formulation is the number of possible solutions as the depth size increases. Any excess nodes that can be evaluated to a one or a zero can result in trivial operations on an otherwise optimal tree solution and can cause the optimization algorithm to stall as it searches an infinite space of solutions of equivalent fitness. A solution to this problem are in penalizing the number of nodes. The addition of a penalty to the objective for the number of nodes used to represent the mathematical expression causes a trade-off between tree size and sum of squared error fitness metric, prioritizing solution with fewer nodes with the same accuracy. The objective with a penalty on the number of nodes is:

$$\min \sum_{i=1}^{n_{data}} (z_i - v_{i1})^2 + \lambda \sum_{n \in N, o \in \mathcal{O}} y_n^o \tag{5}$$

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where λ is the weight penalizing the number of nodes used. The scalar λ is selected close to zero, for this regression, to not impact the quality of the regression based on the sum of squared errors, but still penalize using extraneous nodes. These additions to the objective to limit model complexity and reduce trivial solutions can improve the tractability of these problems, but further evaluation is required to understand which set of constraints facilitates finding accurate solutions quickly without over-restricting the search space.

DEMONSTRATIONS AND COMPUTATIONAL RESULTS

To demonstrate the capabilities of this formulation, we provide two examples of incorporating derivatives in the symbolic regression process. All computations reported below were performed on a Linux machine with a 64-bit Intel Core i7-5600U, at 2.60GHz, using BARON version 21.3.26[22] to solve the MINLP.

Model discrimination via first derivative

Including first derivative constraints can discriminate between various model choices if the function form defined by the data is not unique or if the first derivative must be constrained in order to enforce physically relevant properties. For an illustrative example, we use data points where a line and quadratic polynomial coincide, but the first derivatives can discriminate between the two models. Often first derivative values are not known but there is prior knowledge on the sign of the first derivative based on physical knowledge. For this example, data is sampled from the function $z = x^2 - 2$, which can also be matched by the line $z = -0.25x + 2.5$ at the measured points of $x = \{-2.25, 2\}$. By enforcing sign constraints on the first derivatives, such as:

$$x_i' > 0 \qquad \text{if } x_i > 0 \tag{6}$$

$$x_i' < 0 \qquad \text{if } x_i < 0 \tag{7}$$

at the sampled data points, the MINLP for symbolic regression can find the quadratic functional form, as shown in Figure 2. Additional sampled data points could have resulted in the same regressed model. Enforcing gradient conditions replaces the need for additional measurements and results in an accurate model.

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Constraining extrapolated behavior with partial data sampling

Including first derivative data or constraints can reduce the amount of data points needed in the regression and correct behavior in extrapolated areas. In this example, we use a function with a saddle point, $z = x^2 - y^2$ to evaluate regressions with partial data sets. For this function, we evaluated the regression surfaces obtained using different data sets.

For each regression, we use the same ten randomly sampled data points from $x = [-4, 4]$ and $y = [-4, 4]$. For the one-dimensional sampling of the two-dimensional space, we project the data points onto the x or y axis. An additional three data points are randomly sampled to constrain the first derivative values. Similar to the previous example, it is unlikely that for a new system we will have point-wise derivative information, but we could have knowledge on the derivative sign. The sign constraint on the derivative is applied with respect to the variable y and is enforced by the following equation:

$$y_i' > 0 \qquad \text{if } y_i < 0 \tag{8}$$

$$y_i' < 0 \qquad \text{if } y_i > 0 \tag{9}$$

Each of the regressions converged in less than 60 seconds. Using all the data points made it possible to recover the original equation as shown by Figure 3a. By using only partial data sets with sampling along only one axis, we can represent the parabolic behavior due to one of the variables, as shown in Figures 3b and 3c.

By adding points to constrain the first derivative in the regions away from the sampled axis, we can recover the true function without requiring further evaluated data points to be sampled. We do so by applying knowledge-based constraints on the sign of the derivative as shown in Figure 3d. This ability to constrain the extrapolated behavior in the primary function domain as well as additional derivative domains reduces the need for data points, which could be expensive to measure, and allows us to leverage expert knowledge on a system.

CASE STUDY: CONSTRAINED REGRESSION OF THERMODYNAMIC ALPHA FUNCTION

For a real-world application of our symbolic regression method, we applied our formulation to thermodynamic cubic equations of state. Thermodynamic equations of state relate the state variables, temperature, pressure, and volume, that describe the properties of chemicals from fluids to mixtures. The van der Waals equation,

$$P = \frac{RT}{V_m - b} - \frac{a}{V_m^2} \tag{10}$$

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has a repulsion contribution related to the size of particles and an attraction contribution that are represented by the first and second term, respectively. In Equation (10), V_m is molecular volume, R is the gas constant, T is temperature, b is a constant related to the molecular volume, and a is a function representing the attractive forces.

Since Van der Waals first proposed a functional form for a thermodynamic cubic equation of state (EOS), shown in Equation (10), modifications to the empirical thermodynamic EOS have focused on the temperature dependence of attraction of real fluids. In particular, various cubic equations have been developed over time, such as the Peng-Robinson equation,

$$P = \frac{RT}{V_m - b} - \frac{a_c \alpha(T_r, \omega)}{V_m(V_m + b) + b(V_m - b)} \tag{11}$$

$$a_c = 0.45724 R^2 T_c^2 / P_c$$

$$\alpha(T_r, \omega) = (1 + \kappa(\omega)(1 - T_r^{1/2}))^2$$

$$\kappa(\omega) = 0.37464 + 1.54226\omega - 0.26992\omega^2$$

and the Soave modification of the Redlich-Kwong equation,

$$P = \frac{RT}{V_m - b} - \frac{a_c \alpha(T_r, \omega)}{V_m(V_m + b)} \tag{12}$$

$$a_c = 0.42748 R^2 T_c^2 / P_c$$

$$\alpha(T_r, \omega) = (1 + \kappa(\omega)(1 - \sqrt{T_r}))^2$$

$$\kappa(\omega) = 0.480 + 1.574\omega - 0.176\omega^2$$

These cubic equations of state are widely used in the modeling and optimization of industrial processes. Equation-based flowsheet optimization has been studied specifically for the efficient handling of the thermodynamic calculations using the cubic equations of state[23,24]. Recent work has been conducted to replace the calculation of thermodynamic properties using artificial neural networks but, in addition to requiring a large number of measurements for accurate training, the larger neural networks can exceed problem sizes for commercial solvers and require custom solutions[25,26].

Between the different cubic equations of state, the accuracy is mainly determined by the model for attraction. The attraction contribution of cubic equations of state is formulated with a constant critical attraction, a_c, based on critical temperature and volume and the alpha function, $\alpha(T)$, with a temperature dependence, and is written as:

$$a(T) = a_c \alpha(T_r, \omega) \tag{13}$$

where $\alpha(T)$ is the focus of our regression case study.

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Background on the alpha function

The various models of the alpha function are dependent on parameters such as temperature, density, or acentric factor. For more consistency of equations between multiple species of chemicals, the temperature and density are sometimes used in their reduced form, scaled by critical temperature and critical pressure shown in Equations (14) and (15):

$$T_r = T/T_{crit} \tag{14}$$

$$P_r = P/P_{crit} \tag{15}$$

Modifications to the Peng-Robinson (PR) and Redlich-Kwong (RK) forms of the cubic EOS, commonly used by industry, have focused on the functional form of the alpha function regressed prior to fitting the attraction and repulsion parameters for the cubic equation[27,28]. Traditionally, the parameters of the cubic equations of state are derived from the critical point, which can lead to errors in liquid density calculations and in the critical region. Alternatively, when fit to specific liquid density and vapor pressure data, the equations fail to represent the critical constraints[29]. The latter require that the first and second derivatives of pressure with respect to volume are equal to zero at the critical point.

Rules for the development of alpha functions, defined by Coquelet et al.[30], require that the functions (a) be finite and positive at all temperatures, (b) equal one at the critical point, (c) tend to zero when T goes to infinity, and (d) have continuous primary expression and first and second derivatives. For thermodynamic consistency, Le Guennec et al.[31] recently demonstrated that the alpha functions of cubic equations of state must be continuous, monotonically decreasing ($d\alpha/dT < 0$), convex ($d^2\alpha/dT^2 > 0$), and satisfy ($d^3\alpha/dT^3 < 0$) for any value of temperature. At very high temperatures, the derivatives must converge to zero.

Following the development of these new rules for thermodynamic consistency of alpha functions, there has been further evaluation of commonly used alpha functions within the Redlich-Kwong and Peng-Robinson equations of state in the supercritical regime. Ghanbari et al.[32] compared these two equations over various thermodynamic regions of importance, such as critical and subcritical, with a focus on the alpha function values. The cubic EOS were assessed using PVT calculations in the vapor-liquid equilibrium and Joule-Thomson inversion curves to evaluate their ability to describe critical behavior. Mahmoodi et al.[33,34] developed new alpha equations that adhere to the derivative constraints by using exponential alpha functions and evaluated their behavior in the super critical region. A follow-up by Le Guennec et al.[35] on the imperative need for consistent alpha functions quantified the improved accuracy for supercriticial residual enthalpy calculations at approximately a factor of ten using a consistent alpha function compared to the volume-translated Peng-Robinson equation.

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Derivative constraints on the alpha function

Our application focuses on developing a consistent alpha function by applying the first and second derivative constraints on an alpha function embedded in the Peng-Robinson equation of state, Equation (11). The first derivative of the alpha function with respect to temperature is required to be a monotonically decreasing function. This represents the loss of attraction at higher temperatures caused by increased motion of particles, and follows intuitively with phase transformation of chemicals from liquid to gas where the attraction force felt between particles reduces with increasing temperature.

The behavior of the second derivative of the alpha function with respect to temperature is a more complex thermodynamic function. The second derivative is important for the residual heat capacity and represents the curvature of the alpha function. When the second derivative is zero, the residual contribution to heat capacity is zero and the heat capacity is therefore that of an ideal gas. Multiple inflection points are considered a non-physical solution by Le Guennec et al. and are considered as thermodynamically inconsistent.

Multiple works have sought to evaluate the adherence of existing alpha functions to the new consistency qualifications and to refit the alpha equations. In this work, we use common parts of these alpha functions to generate the input variables as functions of temperature. We refer to these functional expression segments of the alpha functions as motifs. These motifs are summarized in Table 7, where T_c is the critical temperature and T/T_c is the reduced temperature.

For this work, we regressed alpha equations for carbon dioxide and argon in the Peng-Robinson equation of state and compared the equations with the equations presented by Le Guennec et al. We evaluate constrained symbolic regression fit to saturated vapor pressure and saturated liquid isobaric heat capacity[36]. Tree depths of three, four, and five were evaluated with the various constraints and data sets to evaluate the ability of our formulation to identify physically consistent alpha functions with varying complexity. The mathematical operators $+$, $-$, $\times$, $/$, exp, log, $\sqrt{}$, $(\cdot)^2$, and $(\cdot)^3$ are available to construct the alpha function with symbolic regression. For tractability, a penalty was added to the objective to penalize the number nodes. The regression was performed in sequential runs of 1000 seconds. The improved upper bound of a run is used as a cutoff value for the following run to fathom worse solutions. If no improvement occurs during a run of 1000 seconds, a final run is performed for 100,000 seconds.

Results of the regression of the alpha function

The importance of the proposed symbolic regression methodology for the alpha functions is the ability to constrain the derivatives, regress an accurate equation, and identify trends of alpha equations that meet the constraints for thermodynamic consistency. The alpha equations regressed with the first and second derivative constraints follow similar

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patterns that are identified by our formulation. A majority of the alpha functions use exponential functions to achieve the alternating signs of the consecutive derivatives. The common functions when both first and second derivative constraints are applied are shown in Table 8. Without the first and second derivative constraints, the functional forms chosen include simple polynomials and functions with discontinuities due to divisions.

The acentric factor was not used in the functional forms for argon or carbon dioxide. Due to the node penalty used to make this optimization formulation more tractable, the acentric factor for the argon alpha functions is not selected because it is zero. However, for carbon dioxide, since the acentric factor is a constant, it can be replaced by an equivalent constant. A constant input variable could cause further redundant solutions for a functional form and lead to longer solution times. A post regression evaluation of the constants could be used to find any functional relations between the acentric factors of various compounds and constants selected by the regression.

For argon, the regressions using depth three to depth five binary expression trees are shown in Figures 4a to 6a. With increased tree depth and thus functional complexity, the vapor pressure is fit more accurately. The alpha values to fit the vapor pressure data points do not correspond well with the Soave and Twu alpha functions. Ghanbari et al. noted this previously that alpha functions obtained from the Peng-Robinson equation are inconsistent at high temperatures [32]. Correcting this behavior by forcing derivatives to be positive or negative could cause deviations in the fit to the vapor pressure. For higher temperatures, the depth five argon alpha functions have negative alpha values or increasing slopes. To correct this an iterative approach with additional data points could be used for improved high temperature behavior.

For carbon dioxide, the regressions using depth three to depth five binary expression trees are shown in Figures 4b to 6b. The Twu and Soave alpha functions are more consistent with the vapor pressure, again noted by Ghanbari et al. [32]. However, compared to the argon alpha function, there is more extrapolated behavior that violates the thermodynamic constraints applied to the alpha function. At lower temperatures, the extrapolated alpha function has asymptotic behavior for both depths four and five. At higher temperatures, the alpha function can become negative. This highlights a weakness of our current symbolic regression formulation in that it only addresses derivatives at the data points.

Since we use a finite data set over a small range of temperatures, the constraints are only applied to the data set range. Any extrapolated behavior might not be consistent. Deviations at extrapolated data points were previously noted by Keijzer et al. [37] and Koza et al. [1] despite enforcing protected domains for operators, such as division protections against introducing discontinuities, and derivative constraints over the regressed values. To address this issue, an interactive approach was used to add additional data points, where the identified functions violate the constraints, into the regression. These data points do not need output values and are not included in the objective function's sum of squared errors when refitting the equation. By adding additional data points, the constrained behavior of the function can be

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addressed by the symbolic regression. For carbon dioxide, we refit the depth five alpha function with additional data points with results shown in Figure 7. The corrected alpha function uses an exponential function to fit the data and maintain correct derivative behavior in the lower temperature region. In addition to refitting with additional data points for constraint violations, we could refine the function using regular parameter estimation, by making the constants parameters. However, overall, the functions obtained here adhere to the thermodynamic consistency constraints.

CONCLUSION

In this paper, we developed an MINLP methodology for symbolic regression that incorporates derivatives into the binary expression tree structure. With the illustrative examples and thermodynamic case study, we show how derivatives can enhance the regression by allowing constraints for physical behavior and providing additional information to guide the regression. By evaluating the thermodynamic data with constraints, we gained insight into the possible function formulations and necessary complexities, as well as support for the expressions of alpha functions that have been introduced. Further work into the control of complexity of the functions, constraints applying over more than just the basic subtree unit of the parent node and children nodes, and using informational content of inputs and outputs could further improve the tractability and consistency of results of symbolic regression via MINLP.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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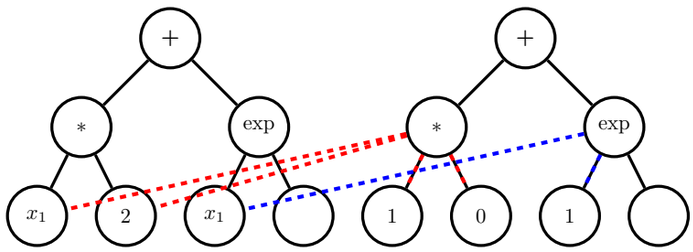


FIGURE 1 Example of the tree representations of the primary function $z = 2x_1 + \exp(x_1)$ and first derivative. The left binary expression tree evaluates the main expression. The right binary expression tree evaluates the first derivative expression with respect to x_1 using values from the main tree nodes and its own nodes. The red and blue dashed lines link nodes used to evaluate the first derivatives for the multiplication and exponential operators, respectively, using the chain rule and values from the primary function and first derivative trees.

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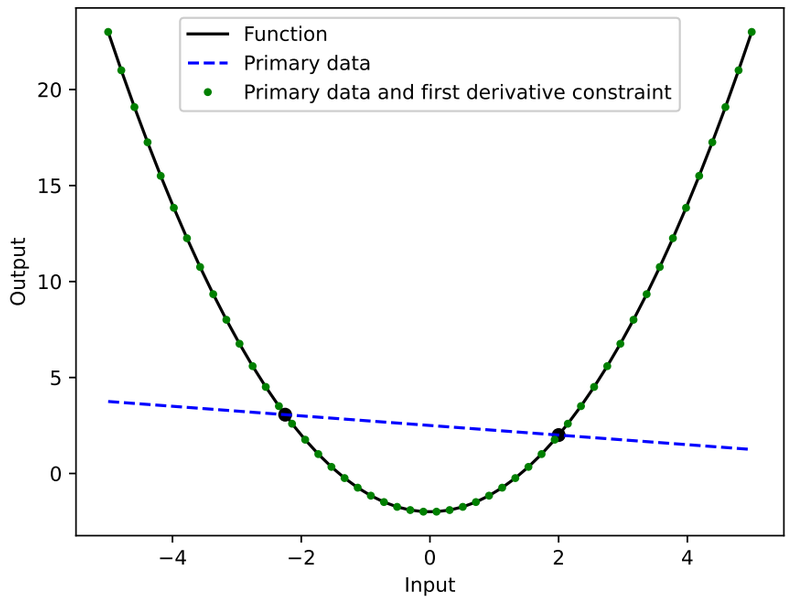
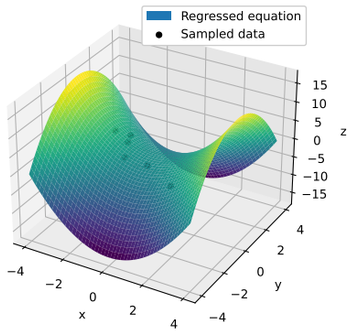


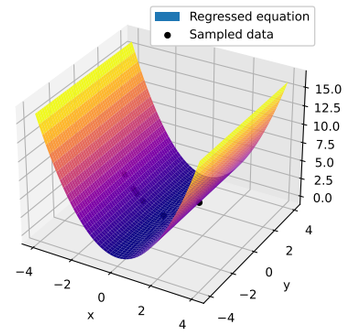
FIGURE 2 Using only the sampled data points, the line is found in blue (- -). Enforcing constraints on the derivatives results in a quadratic functional form shown in green (·).

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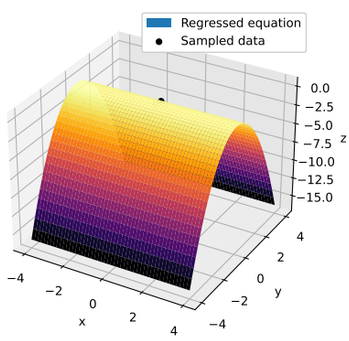
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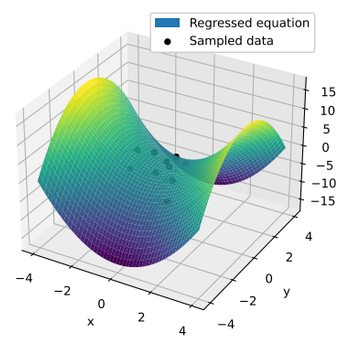
(a) Fully sampled data set of the primary function $z = x^2 - y^2$



(b) Sampled along $y = 0$



(c) Sampled along $x = 0$



(d) Sampled along $y = 0$, with constrained slopes in extrapolated regions

FIGURE 3 Regression surfaces of the function $z = x^2 - y^2$ with data sets of (a) sampled x and y data points, (b) sampled along $y = 0$, (c) sampled along $x = 0$, and (d) sampled along $y = 0$ with constrained slope values in extrapolated regions.

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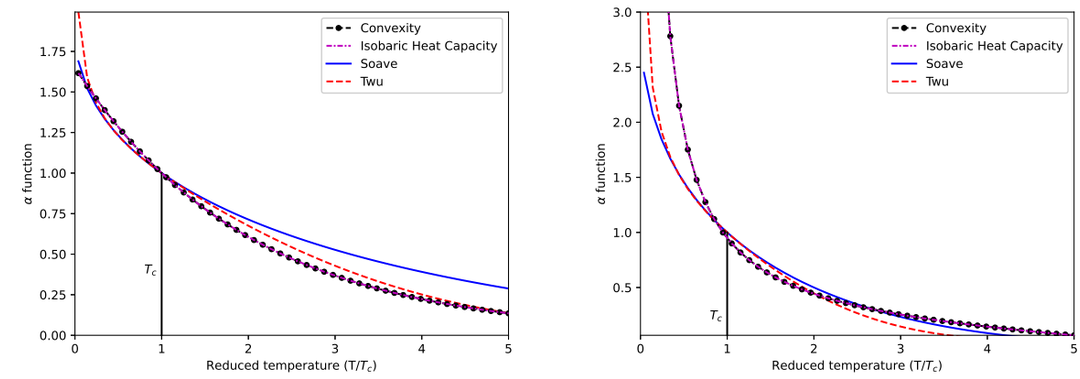


FIGURE 4 Alpha functions with a tree depth of three for (a) Argon and (b) Carbon Dioxide

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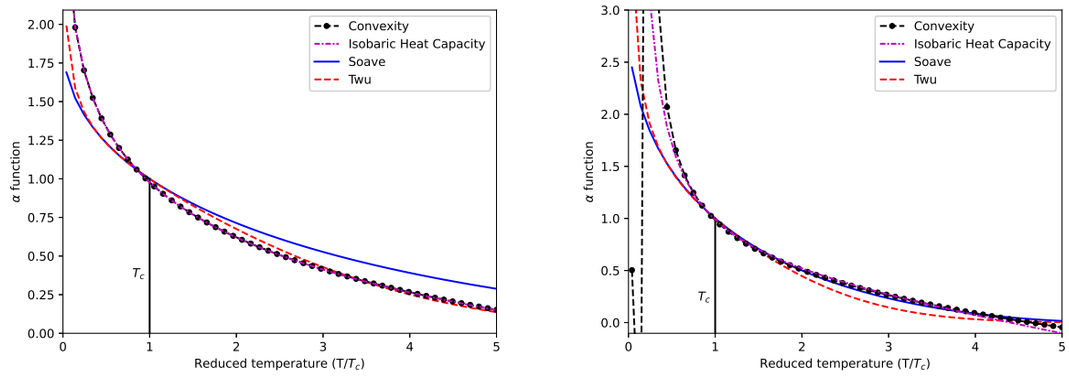


FIGURE 5 Alpha functions with a tree depth of four for (a) Argon and (b) Carbon Dioxide

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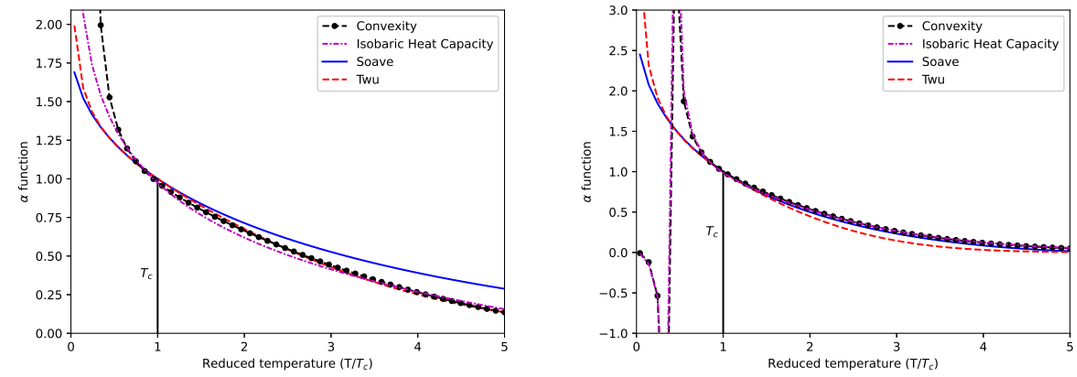


FIGURE 6 Alpha functions with a tree depth of five for (a) Argon and (b) Carbon Dioxide

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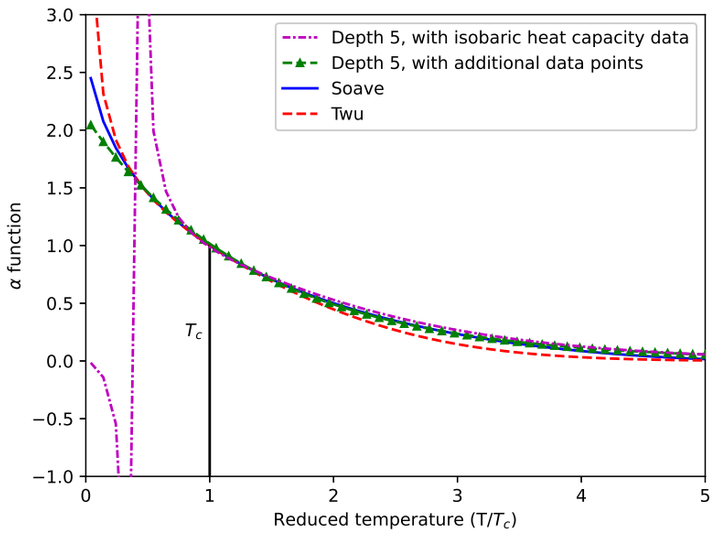


FIGURE 7 Carbon dioxide alpha function refit with additional data points to correct low temperature asymptotic behavior.

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TABLE 1 Basic notation for defining the expression tree

Variable	Description	Defined for elements	Allowed values
Parameters			
z_i	Response z of data point i	$i = 1, ..., n_{data}$	Given
x_{id}	Predictor x_d at data point i	$i = 1, ..., n_{data}$; $d = 1, ..., n_{pred}$	Given
Variables			
v_{in}	Value at the node n and data point i	$i = 1, ..., n_{data}$; $n = 1, ..., N$	$[v_{in}^{lo}, v_{in}^{up}]$
y_n^o	Binary activity variable of operator o at node n	$o \in \mathcal{O}$; $n = 1, ..., N$	$0, 1$
Sets			
$\mathcal{O}$	Superset of simple operators	-	$\mathcal{O} = \mathcal{B} \cup \mathcal{U} \cup \mathcal{L}$
$\mathcal{B}$	Binary operators	-	$+, -, \times, \div, ...$
$\mathcal{U}$	Unary operators	-	$\exp, \log, \sqrt{},...$
$\mathcal{L}$	Leaf node operators	-	$\bigcup_{d=1}^{D} x_d \cup cst$
$\mathcal{V}$	Variable operands	-	$\bigcup_{d=1}^{D} x_d$
$\mathcal{T}$	Terminal Nodes	-	$\{n; d = D\}$

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TABLE 2 Common operand active node functions

Description	Binary	$f_i^0 = 0$	$g_i^0 <= 0$
Binary operations			
Addition	y_n^+	$v_{il_n} + v_{ir_n} - v_{in}$	
Subtraction	y_n^-	$v_{il_n} - v_{ir_n} - v_{in}$	
Multiplication	$y_n^\times$	$v_{il_n} v_{ir_n} - v_{in}$	
Division	$y_n^\div$	$v_{il_n} - v_{in} v_{ir_n}$	$\epsilon - (v_{ir_n})^2, \epsilon - (v_{il_n})^2$
Unary operations			
Exponential	$y_n^{\exp}$	$\exp(v_{il_n}) - v_{in}$	
Logarithm	$y_n^{\log}$	$v_{il_n} - \exp(v_{in})$	$\epsilon - v_{il_n}$
Square	y_n^{sqr}	$(v_{il_n})^2 - v_{in}$	
Cube	y_n^{cube}	$(v_{il_n})^3 - v_{in}$	
Square root	y_n^{sqrt}	$v_{il_n} - (v_{in})^2$	$\epsilon - v_{il_n}$
Leaf node operands			
Variable	y_n^{var}	$x_{id} - v_{in}$	
Constant	y_n^{const}	$v_{i'n} - v_{in}, \forall i \neq i'$	

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TABLE 3 First Derivative Notation

Description	Binary	$f_i'^o = 0$
Binary operations		
Addition	y_n^+	$v'_{il_n} + v'_{ir_n} - v'_{in}$
Subtraction	y_n^-	$v'_{il_n} - v'_{ir_n} - v'_{in}$
Multiplication	$y_n^\times$	$v'_{il_n} v_{ir_n} + v_{il_n} v'_{ir_n} - v'_{in}$
Division	$y_n^\div$	$(v'_{il_n} - v_{in} v'_{ir_n})/v_{ir_n} - v'_{in}$
Unary operations		
Exponential	$y_n^{\exp}$	$v'_{il_n} \exp(v_{il_n}) - v'_{in}$
Logarithm	$y_n^{\log}$	$v'_{il_n}/v_{il_n} - v'_{in}$
Square	y_n^{sqr}	$2v'_{il_n}(v_{il_n}) - v'_{in}$
Cube	y_n^{cube}	$3v'_{il_n}(v_{il_n})^2 - v'_{in}$
Square root	y_n^{sqrt}	$0.5v'_{il_n}/(v_{il_n})^{1/2} - v'_{in}$
Leaf node operands		
Variable	y_n^{var}	$1 - v'_{in}$
Constant	y_n^{const}	$0 - v'_{in}$

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TABLE 4 Second Derivative Notation

Description	Binary	$f_i''^{o} = 0$
Binary operations		
Addition	y_n^{+}	$v''_{il_n} + v''_{ir_n} - v''_{in}$
Subtraction	y_n^{-}	$v''_{il_n} - v''_{ir_n} - v''_{in}$
Multiplication	$y_n^{\times}$	$v''_{il_n} v_{ir_n} + 2v'_{il_n} v'_{ir_n} + v_{il_n} v''_{ir_n} - v''_{in}$
Division*	$y_n^{\div}$	$(v''_{il_n} - 2v'_{ir_n} v'_{in} - v_{in} v''_{ir_n})/v_{irn} - v''_{in}$
Unary operations		
Exponential	$y_n^{\exp}$	$v''_{il_n} \exp(v_{il_n}) + (v'_{il_n})^2 \exp(v_{il_n}) - v''_{in}$
Logarithm	$y_n^{\log}$	$v''_{il_n}/v_{il_n} - (v'_{il_n}/v_{il_n})^2 - v''_{in}$
Square	y_n^{sqr}	$2(v''_{il_n} v_{il_n} + (v'_{il_n})^2) - v''_{in}$
Cube	y_n^{cube}	$3v''_{il_n} (v_{il_n})^2 + 6(v'_{il_n})^2 v_{il_n} - v''_{in}$
Square root	y_n^{sqrt}	$0.5v''_{il_n}/(v_{il_n})^{1/2} - 0.25(v'_{il_n})^2/(v_{il_n})^{3/2} - v''_{in}$
Leaf node operands		
Variable	y_n^{var}	$0 - v''_{in}$
Constant	$y_n{}^{const}$	$0 - v''_{in}$

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TABLE 5 Search space reducing constraints

Space reduced	Example	Source
Commutative property symmetry breaking	$A + B = B + A$	Cozad and Sahinidis[15]
Nested operators exp-log	$\exp(\log(x))$, $\log(\exp(x))$	Cozad and Sahinidis[15]
Nested operators sqrt-sqr	$\sqrt{x^2}$, $(\sqrt{x})^2$	Cozad and Sahinidis[15]
Associative property with constant $(+, -)$	$x + (-1) = x - 1$	Cozad and Sahinidis[15]
Associative property with constant $(\times, \div)$	$x/4 = 0.25 * x$	Cozad and Sahinidis[15]
Binary operation on constants	$1 + 2$, $1 * 2$	Cozad and Sahinidis[15]
Unary operation on constants	$\exp(1)$, $\sqrt{3}$	Cozad and Sahinidis[15]
Multilevel addition/subtraction	$A - (B - C) = A + (C - B)$	Kim et al.[19]
Multilevel multiplication/division	$A/(B/C) = A * (C/B)$	Kim et al.[19]
Implication cuts	$\exp(A) \in \mathbb{R}^+$	Kim et al.[19]
Trivial subtraction of variables	$x_1 - x_1 = 0$	This work
Trivial division of variables	$x_1/x_1 = 1$	This work
Equivalent addition of variables	$x_1 + x_1 = 2 * x_1$	This work
Equivalent multiplication of variables	$x_1 * x_1 = x_1^2$	This work
Multilevel logarithm	$\log(A * B) = \log(A) + \log(B)$	Not constrained
Multilevel exponential	$\exp(A * B) = A' \exp(B)$	Not constrained

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TABLE 6 Illustrative example results

Data Inputs	Regressed Expression	Regression Time (s)
x	$-0.25x + 2.5$	8.31
x, $h(x')$	$x^2 - 2$	45.63

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TABLE 7 Alpha function motifs

Variable	Functional form	Motif source examples
x_1	T/T_c	Common reduced form
x_2	$1 - T/T_c$	[38],[39]
x_3	w	acentric factor
x_4	$1 - \sqrt{T/T_c}$	[40],[41],[42],[43]

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TABLE 8 Example regressed alpha functions for carbon dioxide and argon. Equations are simplified.

Formulation	Expression	Training Error	Test Error
Exponential models			
Argon, Depth 3, with C_p*	$\exp(1-T_r)^{0.5}$	0.0080	0.0365
Argon, Depth 5	$\exp(0.06984/T_r^2) * (0.1468T_r - 1.1013)^2$	0.0017	0.0060
CO_2, Depth 5	$1.7301\exp(-1.041T_r)T_r^2/(T_r - 0.3898)$	0.00032	0.000204
CO_2, Depth 5, with C_p	$\exp(1-T_r) * (0.5596T_r + 0.0310)T_r/(T_r - 0.40008)$	0.00032	1.416E-5
CO_2, Depth 5, Corrected	$\exp(0.744 - 0.7284T_r)$	0.00516	0.002878
Other models			
Argon, Depth 4, with C_p*	$0.9768 + (-0.51256 * \ln(T_r))$	0.0039	0.0155
Argon, Depth 5, with C_p	$0.9576 - \ln((T_r - 0.03279)^{0.5})$	0.0039	0.0149
CO_2, Depth 3, with C_p*	$0.9490/T_r + 0.03085 * (1 - T_r)$	0.0064	0.0038
CO_2, Depth 4	$0.3757 * (1 - \sqrt{T_r}) + (\sqrt{T_r} - 0.2374)/(T_r - 0.2239)$	0.000664	0.000417
CO_2, Depth 4, with C_p	$-0.1427T_r + 0.649/T_r + 0.4814$	0.000686	0.000424

* Same equation is found using only vapor pressure and both vapor pressure and isobaric heat capacity

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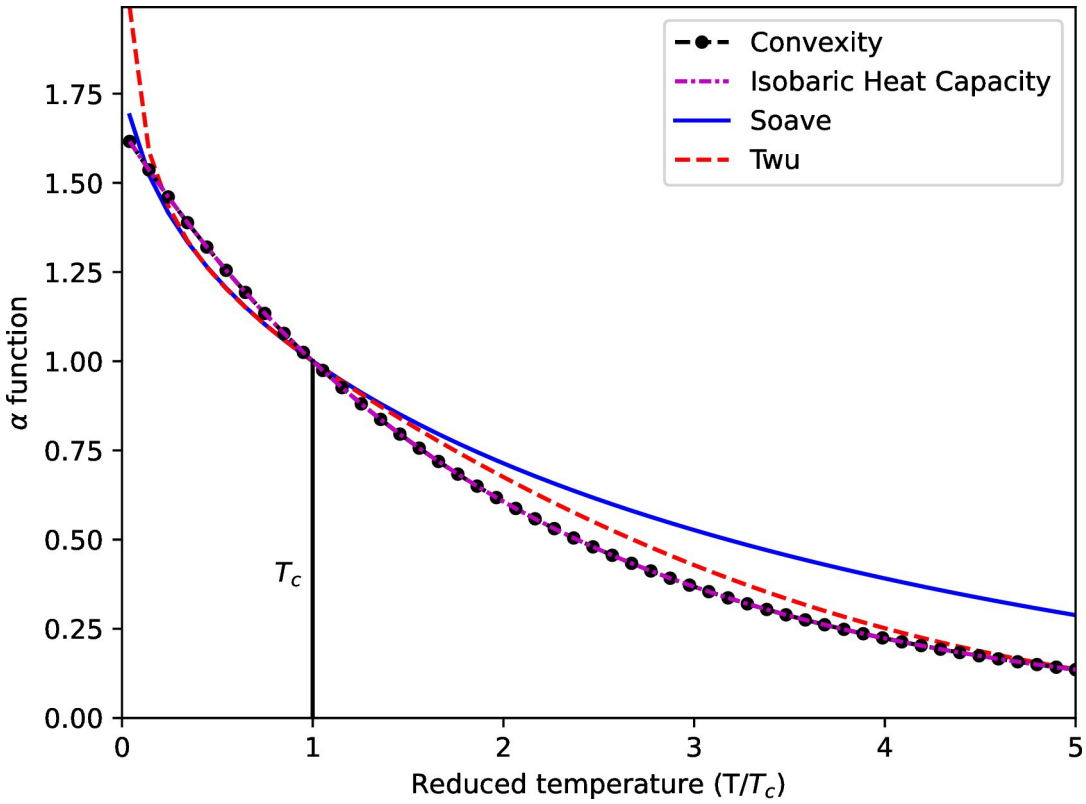
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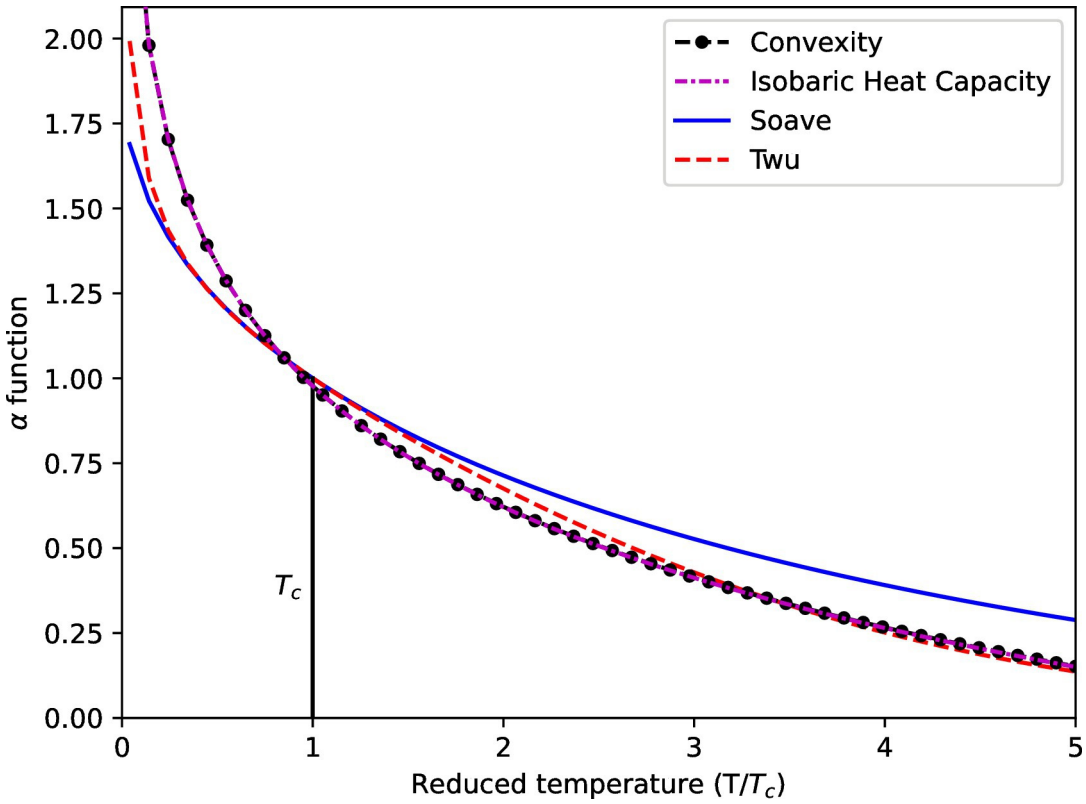
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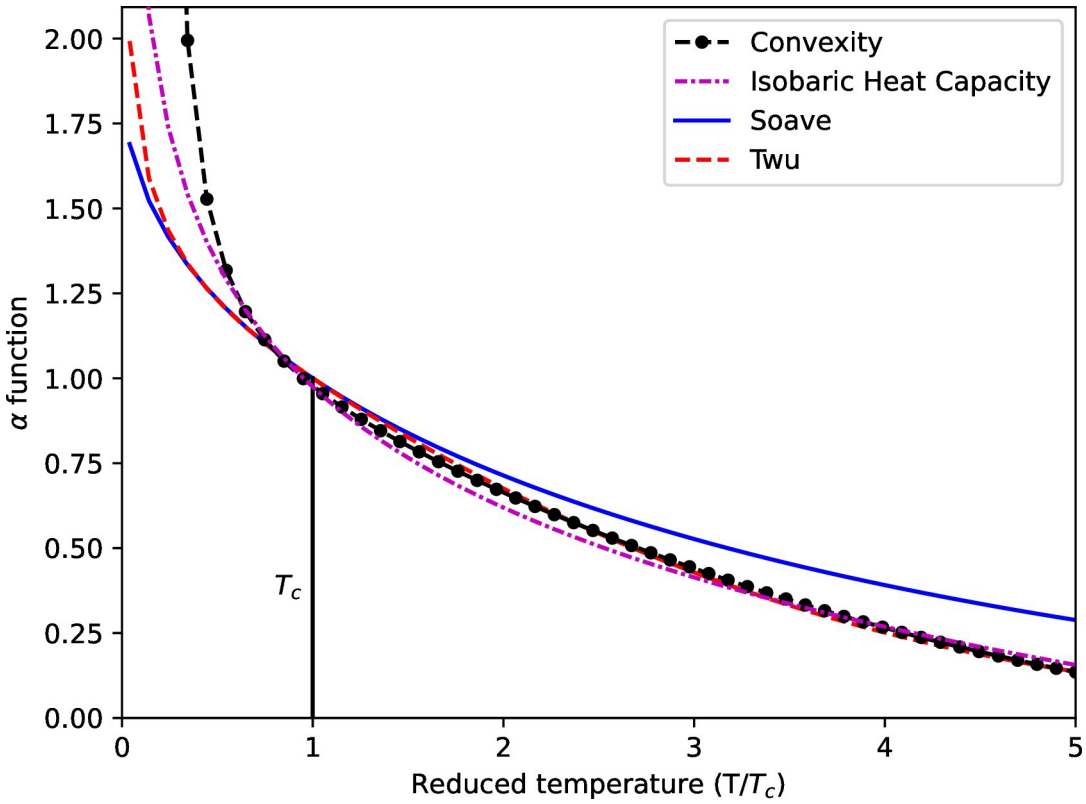
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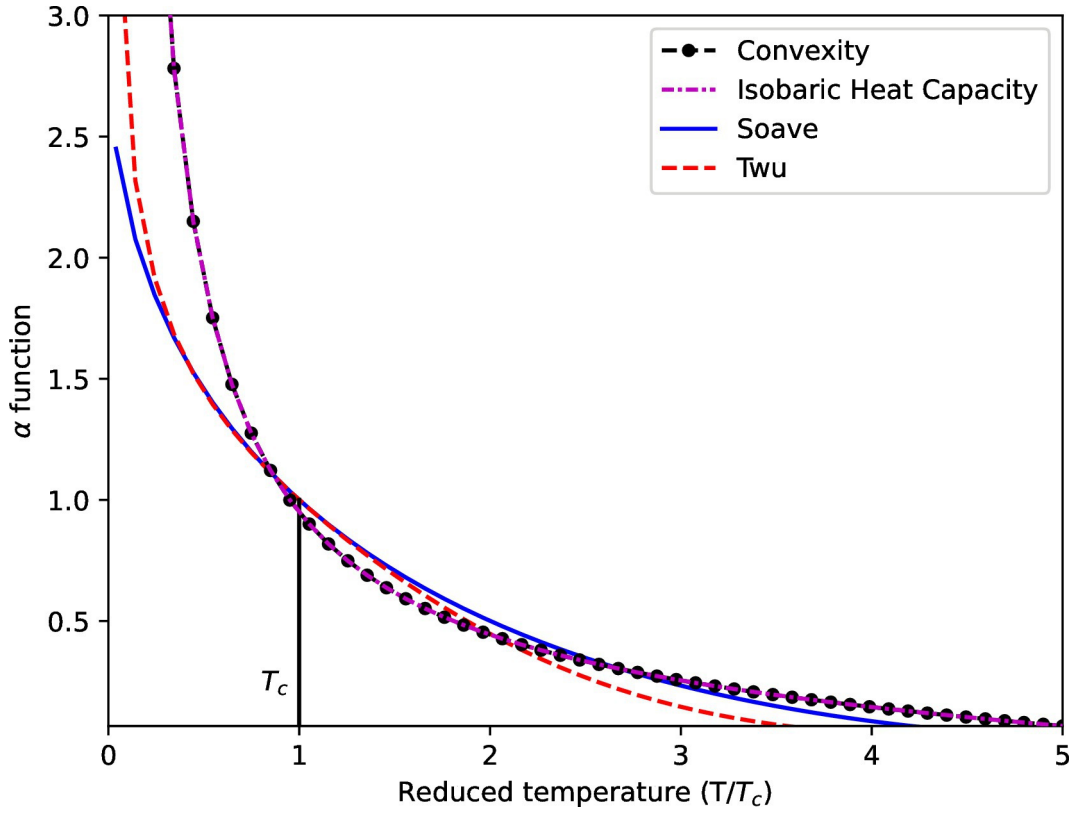
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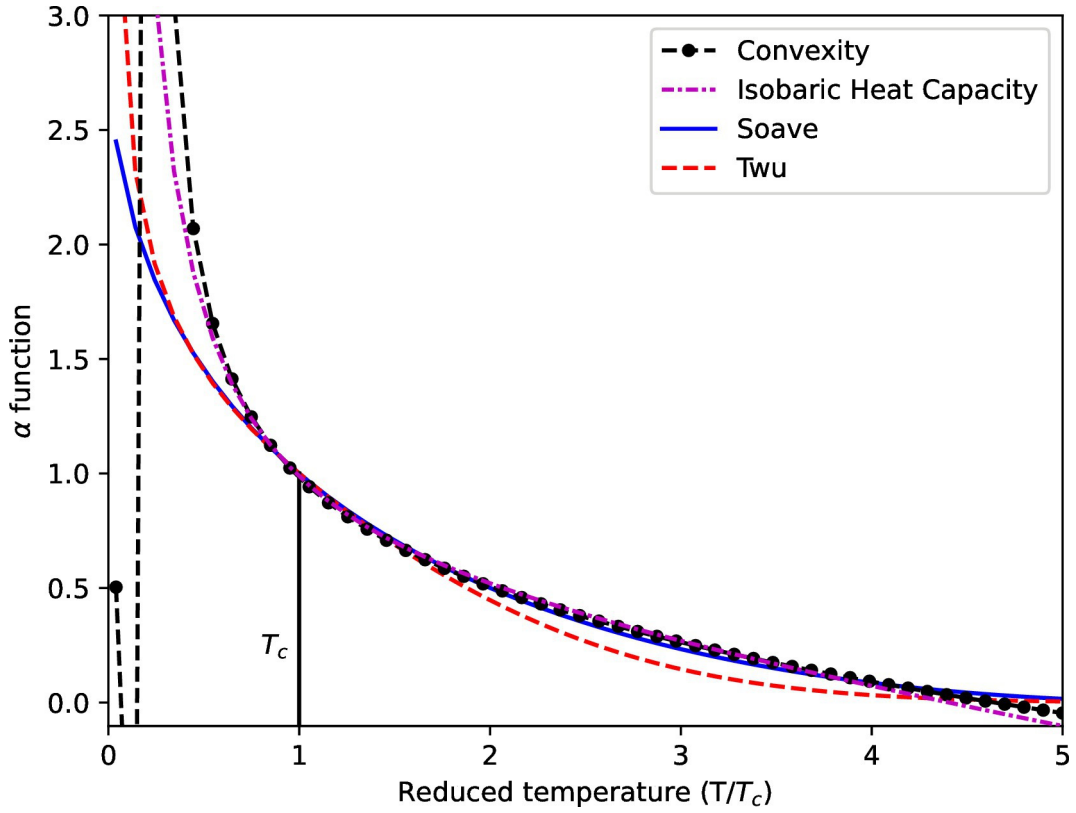
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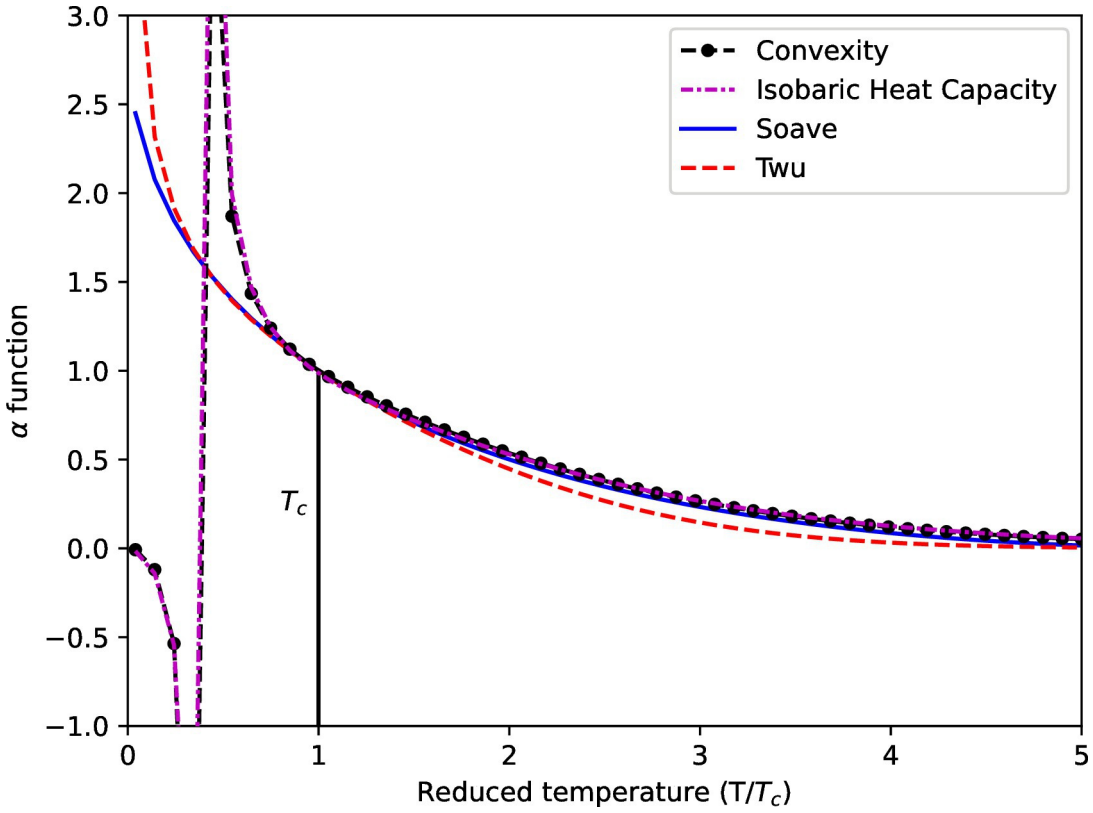
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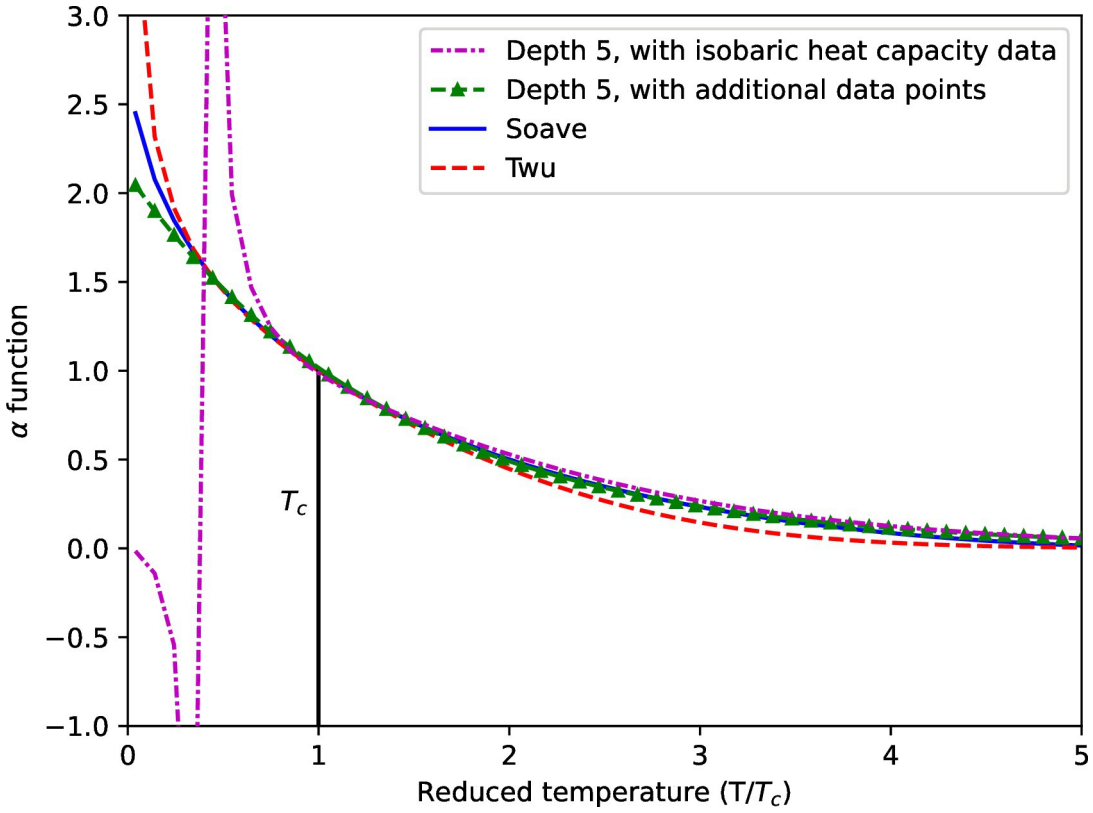
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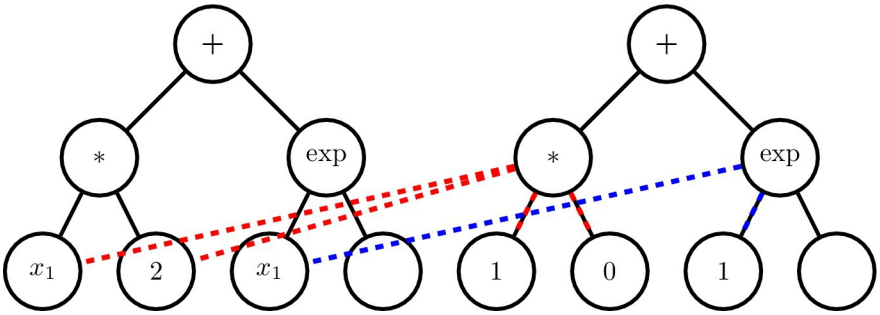
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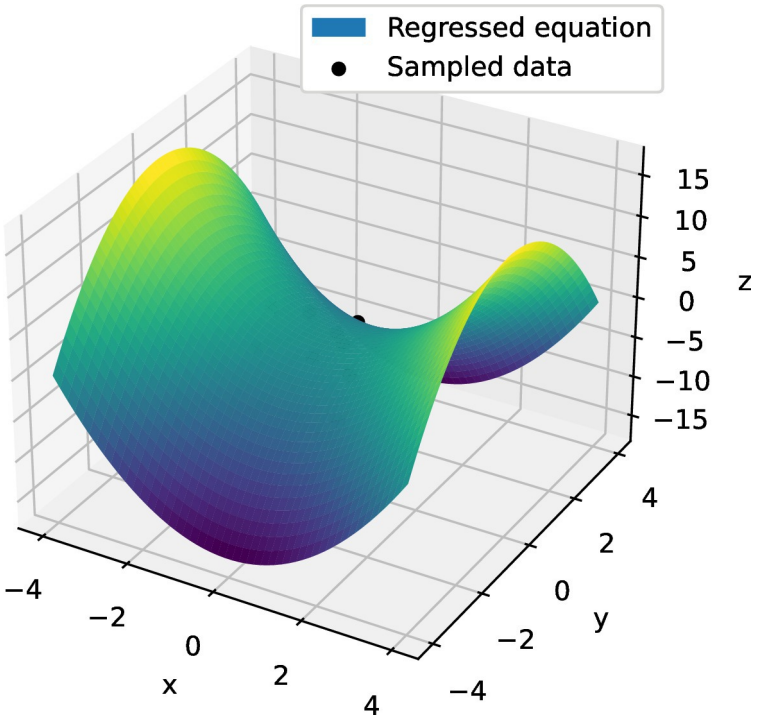
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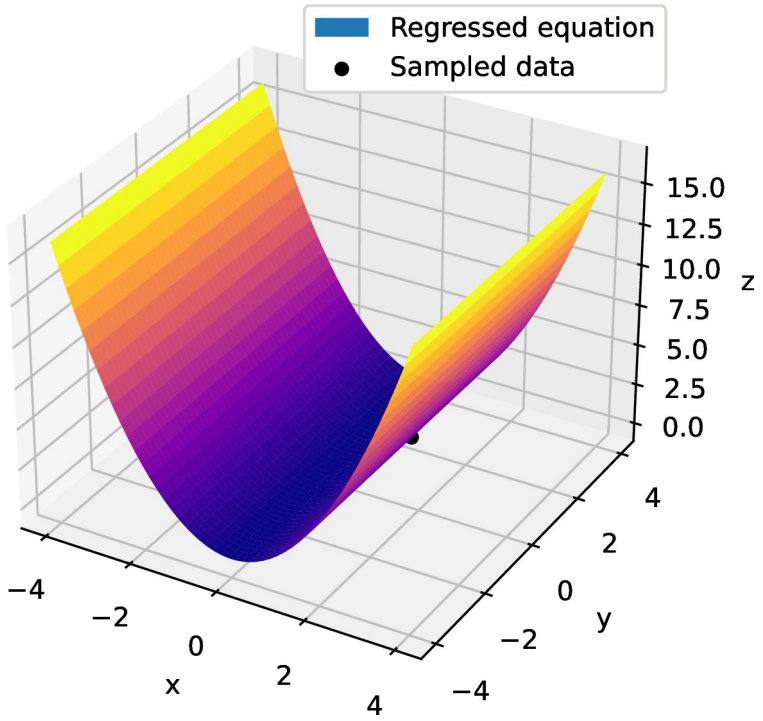
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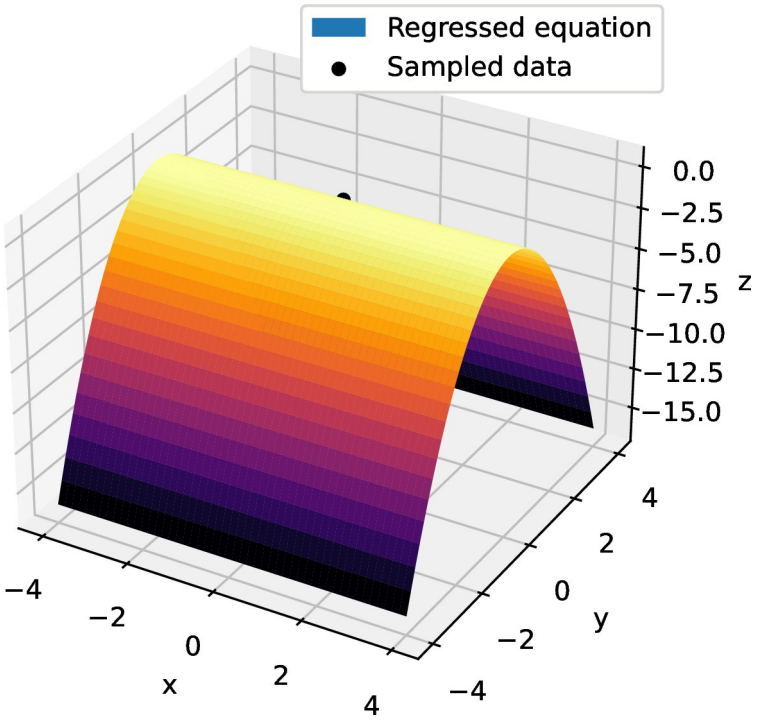
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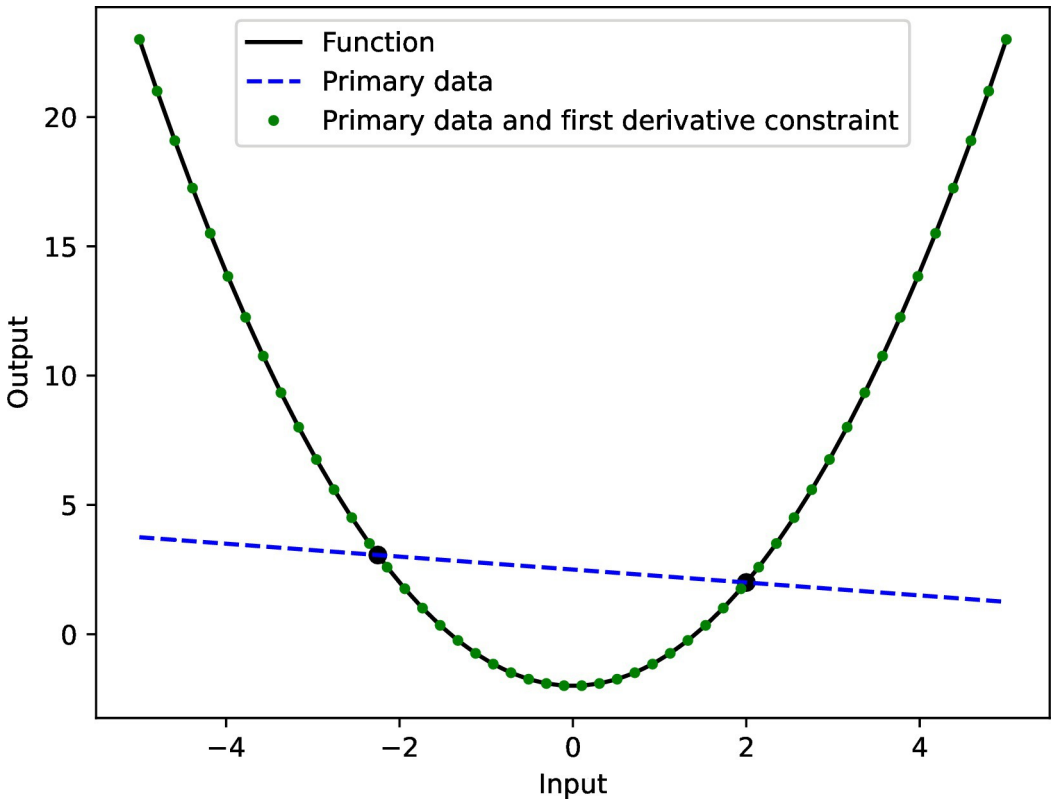
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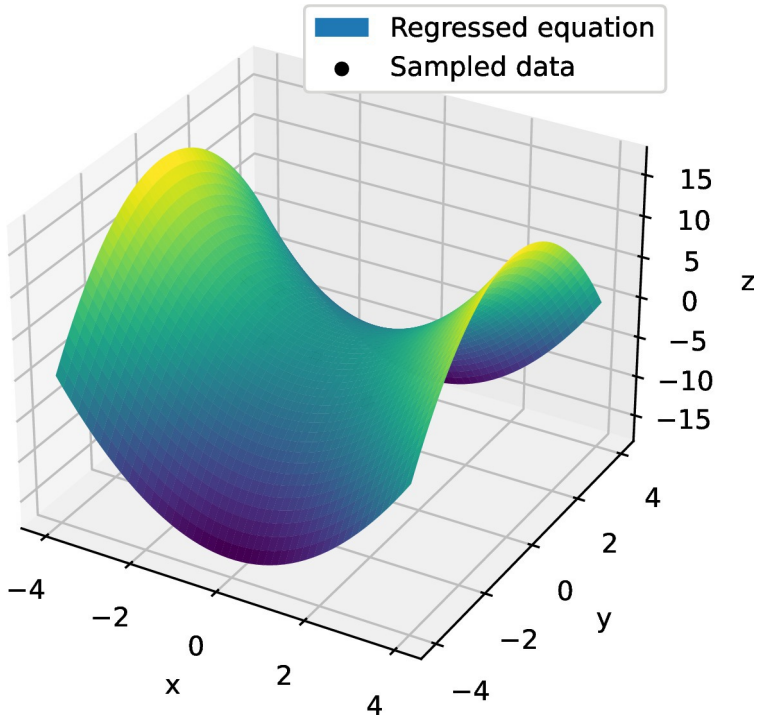
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